

Qn	Suggested solution	Remarks
1(a)	Method 1: Using equations of motion Taking downwards to be positive: $v^2 = u^2 + 2as = 1.5^2 + 2(9.81)(0.65)$ $v = \pm 3.9 \text{ m s}^{-1}$ Final speed = 3.9 m s⁻¹ OR Method 2: Using conservation of energy Loss in GPE = Gain in KE $mgh = \frac{1}{2}m(v^2 - u^2)$ $v^2 = 2gh + u^2$ $v = 3.9 \text{ m s}^{-1}$ Final speed = 3.9 m s⁻¹	[1] sub [1] ans
MC		
(b)	velocity / m s⁻¹ time [1] Straight lines from X to t_1 and t_1 to Y; and both with negative gradients. [1] The two straight lines must be parallel and with correct values (ecf).	
(c)	The speed of the ball just after rebound is less than the speed just before impact. Hence there is a loss/change in the kinetic energy of the system during the bounce. The bounce is inelastic.	[1] [1]
MC	State without reason, [0] . State with loss/change in kinetic energy [1] .	
(d)	Graph for downward motion - smaller gradient and duration is longer than t_1 Graph for upward motion - larger gradient and duration is shorter than $(t_2 - t_1)$	[1] [1]
2(a)	1. The vector sum of all external forces acting on a body must be zero	[1]